

TSA Meeting — Slide 11 Study Note

Ch 5: Adaptive Distance Protection — Zone Compensation for IBR

Contributions C5 · Chapter 5 · SAMBP Framework

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Slide Overview and Narrative

What This Slide Shows

Slide 11 presents **Ch 5: Adaptive Distance Protection**, covering Contributions C5 and C6. The slide's narrative has three parts:

The problem: Distance relays measure apparent impedance Z_{app} . IBR reactive current injection during a fault ride-through adds a complex shift $\Delta Z_{ibr}(k_{ibr}, \varphi_{ctrl})$ to the measured impedance. For lagging reactive injection (the dominant Volt-VAr control mode during voltage dips), the shift pushes Z_{app} *into* Zone 1 for faults beyond 80% of the line — causing **overreach** (a false Zone 1 trip). For leading injection, the shift pushes Z_{app} *out* of Zone 1 — causing **underreach** (a missed trip).

The solution: The SAMBP-21 relay implements two complementary mechanisms: (1) an **adaptive quadrilateral Zone 1 boundary** that corrects for ΔZ_{ibr} using the LM-estimated $\hat{k}_{ibr}$, updated every 10 ms; and (2) **Zone 2 POTT acceleration** that bypasses the 300 ms Zone 2 timer when $k_{ibr} > 0.5$, clearing faults within 32 ms via GOOSE permissive signalling.

The result: Overreach events fall from 17/50 to 2/50 (from 34% to 4%), underreach from 5/50 to 0/50, correct Zone 1 trips rise from 74% to 96%, with only 1 ms additional average trip time.

Slide Key Results

Metric	Fixed Relay	SAMBP-21	Change
Overreach events	17/50 (34%)	2/50 (4%)	−88% relative
Underreach events	5/50 (10%)	0/50 (0%)	eliminated
Correct Zone 1 trip	74%	96%	+22 pp
Avg trip time	18 ms	19 ms	+1 ms only

Key observation from slide: Zone adaptation nearly eliminates both overreach and underreach simultaneously, with negligible speed penalty. This is the central claim of Contribution C5.

Engineering Challenge: Why Fixed Zone 1 Fails for IBR

The Core Problem: Apparent Impedance Distortion

A distance relay at End A of a transmission line measures:

$$Z_{\text{app}} = \frac{\hat{V}_A}{\hat{I}_A}$$

In a classical synchronous network, for a fault at fractional position α along the line with fault impedance Z_F :

$$Z_{\text{app}} = \alpha Z_L + Z_F \left(1 + \frac{\hat{I}_B}{\hat{I}_A} \right)$$

where $\hat{I}_B$ is the infeed current from the remote end. This is the classical infeed equation — the relay always sees a *larger* apparent impedance than the fault distance alone, never a smaller one.

Why IBR breaks this: A Type-4 IBR under Volt-VAr control injects reactive current $\hat{I}_B = k_{\text{ibr}} I_r e^{j\beta}$ during the fault. The commanded angle β depends on the control mode:

- **Volt-VAr absorb** (most common during nearby faults: V_{PCC} dips): $\beta = -90^\circ$ (lagging) $\Rightarrow Z_{\text{app}}$ shifts *downward* in the X direction $\Rightarrow$ faults at $\alpha > 0.8$ enter Zone 1 $\Rightarrow$ **overreach**.
- **Volt-VAr inject** (remote faults where V_{PCC} rises): $\beta = +90^\circ$ (leading) $\Rightarrow Z_{\text{app}}$ shifts *upward* $\Rightarrow$ Zone 1 boundary missed $\Rightarrow$ **underreach**.

The conventional relay's Zone 1 boundary is set at 80–85% of line length and is *fixed*. It has no knowledge of the current IBR operating point and cannot adapt.

Magnitude of the Problem

For a 100 km, 110 kV line with $Z_L = 0.35 + j0.40 \Omega/\text{km}$ and a fault at $\alpha = 0.85$ (just outside Zone 1 in a classical network):

$$|\Delta Z_{\text{ibr}}| = Z_F k_{\text{ibr}} \frac{I_r}{I_A} = 10 \Omega \times 0.5 \times 1.0 = 5 \Omega$$

The Zone 1 reactive reach is $X_{\text{reach}} \approx 0.80 \times 0.40 \times 100 = 32 \Omega$. A shift of -5Ω in X moves the apparent impedance from $X_{\text{app}} = 34 \Omega$ (outside Zone 1) to $X_{\text{app}} = 29 \Omega$ (inside Zone 1) — a clear overreach. This is not a corner case; at $k_{\text{ibr}} = 0.5$ and $\beta = -90^\circ$, **every fault at $\alpha > 0.8$ overreaches**.

IBR Apparent Impedance Shift: Full Derivation

Complete Apparent Impedance Equation

The apparent impedance with IBR reactive current injection decomposes into three terms:

$$Z_{\text{app}} = \alpha Z_L + Z_F + \underbrace{Z_F k_{\text{ibr}} \frac{I_r}{I_A} e^{j\beta}}_{\Delta Z_{\text{ibr}}(k_{\text{ibr}}, \beta)} \quad (1)$$

The three terms represent:

1. αZ_L : the line impedance to the fault (the correct measurement).
2. Z_F : the fault resistance seen by End A (present in classical networks too).
3. $\Delta Z_{\text{ibr}} = Z_F k_{\text{ibr}} (I_r/I_A) e^{j\beta}$: the IBR infeed contribution (new in IBR networks).

The reactive component of the shift is:

$$\Delta X_{\text{ibr}} = Z_F k_{\text{ibr}} \frac{I_r}{I_A} \sin \beta \quad (2)$$

Slide notation: The slide writes $Z_{\text{app}} = Z_{\text{fault}} + \Delta Z_{\text{ibr}}(k_{\text{ibr}}, \varphi_{\text{ctrl}})$ where φ_{ctrl} corresponds to β (the commanded reactive angle). The “radially outward shift” mentioned on the slide refers specifically to the $\beta = -90^\circ$ case, where ΔZ_{ibr} is a downward vector in the R - X plane, moving the locus toward the origin (into Zone 1).

Why the Shift is Radially Outward from the Relay Perspective

The relay at End A is looking *into* the line (toward increasing αZ_L). “Radially outward” in the relay’s impedance plane means toward larger $|Z|$ — which corresponds to a fault appearing *farther* away. However, for the lagging case ($\beta = -90^\circ$), $\Delta X < 0$, which means Z_{app} moves toward smaller $|Z|$ — i.e., the fault appears *closer* than it is. This is the **overreach** scenario. The phrase “radially outward” in the slide caption refers to the direction of the IBR shift vector in the complex plane (radially from the fault impedance point), not toward a larger apparent distance.

Quadrilateral Distance Characteristic (Contribution C5)

Why Quadrilateral Instead of Mho

The classical Mho (admittance circle) characteristic has two critical limitations:

1. **Zero resistive reach at origin:** The Mho circle passes through the origin. For high-resistance faults ($R_F > 0$), the fault locus runs along the R -axis — outside the Mho circle. The Mho misses resistive faults entirely.
2. **Reactive reach not independently controllable:** Extending the circle to cover resistive faults also extends the X reach, increasing overreach risk.

The **quadrilateral characteristic** provides independent control:

$$C_X : X_1 \leq X_{\text{relay}} \leq X_{\text{reach}} \quad (3)$$

$$C_R : R_1 \leq R_{\text{relay}} \leq R_{\text{reach}} \quad (4)$$

$$C_{\text{load}} : R_{\text{relay}} \geq \tan(\angle Z_L - \delta_{\text{load}}) \cdot X_{\text{relay}} \quad (5)$$

The relay trips when all three conditions are simultaneously met: $T_{21} = \mathbf{1}[C_X \cap C_R \cap C_{\text{load}}]$.

The load blinder C_{load} (set at $\delta_{\text{load}} = 20^\circ$) ensures the operating zone does not overlap with the load impedance region (which also lies in the first quadrant of the R - X plane at high $|Z|$).

IBR Infeed Amplification and Resistance Reach Setting

With IBR infeed, the apparent *resistance* is amplified:

$$R_{\text{app}} = R_F \left(1 + \frac{1}{k_{\text{ibr}}} \right) \quad (6)$$

For $k_{\text{ibr}} = 0.1$ and $R_F = 0.004$ pu (arcing fault): $R_{\text{app}} = 0.004 \times 11 = 0.044$ pu. The quadrilateral forward resistance reach must therefore be set to:

$$R_{\text{fwd},1} \geq R_F^{\text{max}} \left(1 + \frac{1}{k_{\text{ibr}}^{\text{min}}} \right)$$

For $k_{\text{ibr}}^{\text{min}} = 0.10$: $R_{\text{fwd},1} = 0.030$ pu covers all arc faults with $R_F \leq 0.004$ pu at any IBR penetration level $k_{\text{ibr}} \geq 0.10$ (TR-34).

Contrast with Mho: At the Zone 1 circle boundary, the Mho has maximum resistance reach of approximately $R_{\text{mho}} = X_{\text{reach}} \cdot \tan(\angle Z_L)^{-1}$. For $\angle Z_L = 70^\circ$: $R_{\text{mho}} \approx 0.36 X_{\text{reach}}$. For arcing faults with high R_F and IBR amplification, the Mho reach is exhausted before the quadrilateral.

Adaptive Zone Compensation Mechanism

SAMBP-21 Zone Boundary Update Law

The Zone 1 reactive reach is adapted in real time:

$$X_{\text{reach,adapt}}(t) = X_{\text{reach,nom}} - k_X \cdot \hat{k}_{\text{ibr}}(t) \cdot \frac{I_{\text{rated}}}{I_A} \quad (7)$$

where:

- $X_{\text{reach,nom}}$ = nominal Zone 1 reach (0.040 pu, set at 80% of line reactance).
- $k_X = |Z_F^{\text{max}}| \cdot |\sin \beta_{\text{max}}| \approx 10 \Omega \cdot 1 = 10 \Omega$: reactive correction coefficient.
- $\hat{k}_{\text{ibr}}(t)$ = real-time IBR penetration estimate from the Levenberg–Marquardt digital twin (Chapter 6).
- Update period: every **10 ms** (one power-frequency half-cycle).

The correction subtracts the expected reactive shift before it causes overreach, rather than detecting the overreach after the fact.

Zone 2 Acceleration Logic

When the LM estimator detects $\hat{k}_{\text{ibr}} > 0.5$, the IBR contribution is so dominant that Zone 1 instantaneous tripping is no longer reliable even with adaptation. SAMBP-21 then activates **Zone 2 POTT acceleration**:

- The normal Zone 2 timer (200–300 ms) is bypassed.

- A permissive GOOSE signal is sent to End B via IEC 61850.
- End B confirms Zone 2 qualification and directional element *forward*.
- Instantaneous trip is issued at End B.
- End-to-end clearing: $4 \text{ ms} + t_{CT} + t_{\text{meas}} \leq 8 \text{ ms}$ (one cycle).

This is the **Permissive Overreaching Transfer Trip (POTT)** scheme. The 8 ms POTT trip time is well within the 100 ms memory polarisation window ($T_{\text{mem}} = 100 \text{ ms}$), ensuring directional integrity is maintained throughout.

IBR-Secure POTT: Preventing Spurious Permissives

A naive POTT can send a false permissive during a through-fault if IBR injection causes Zone 2 overreach. The IBR-secure POTT adds a k_{ibr} inhibit:

$$T_{\text{POTT}} = T_{Z2,A} \cap T_{Z2,B} \cap \mathbf{1}[k_{\text{ibr}} < 0.45] \quad (8)$$

For $k_{\text{ibr}} > 0.45$, the POTT reverts to Zone 2 with reach reduced to 110% and timer set to 100 ms — sacrificing some speed but maintaining security against through-fault overreach.

Cross-Polarised Mho Element for Directional Security

The SAMBP-21 uses a cross-polarised mho element for the directional supervisory function. The polarising voltage is derived from a healthy phase to eliminate memory-voltage dependency:

$$\hat{V}_{\text{pol}} = \hat{V}_b - \hat{V}_c \quad (\text{for phase A fault}) \quad (9)$$

The operate condition:

$$\text{Re} \left[\hat{I}_a \overline{\hat{Z}_{\text{reach}}} \cdot \hat{V}_{\text{pol}} \right] \geq 0 \quad (10)$$

Because $\hat{V}_{bc}$ remains near rated during single-phase and double-phase faults, the directional element does not lose polarisation as the memory voltage decays. This eliminates failure mode F6 (directional loss after memory expiry, which occurs at $t > T_{\text{mem}}$ for classical self-polarised Mho).

Timing margin: Zone 1 trip time is $t_{Z1} \approx 20 \text{ ms}$. Memory window is $T_{\text{mem}} = 100 \text{ ms}$. Margin: $T_{\text{mem}}/t_{Z1} = 5\times$ — the directional element remains valid even if the Mho element is delayed five-fold (e.g., due to CT saturation or communication delays).

Results: What the Numbers Mean

Interpreting the 50-Fault Simulation Results

The 50-fault simulation covers the full $(\alpha, k_{\text{ibr}}, \beta)$ operating space:

- $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.85, 0.95\}$ (fault location)
- $k_{\text{ibr}} \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ (IBR penetration)
- $\beta \in \{-90^\circ, 0^\circ, +90^\circ\}$ (control mode)

Metric	Fixed Relay	SAMPB-21	Improvement
Overreach events	17 (34%)	2 (4%)	−88%
Underreach events	5 (10%)	0 (0%)	−100%
Correct Zone 1 trip	28 (56%)	48 (96%)	+40 pp
Zone 2 clearance	22 (44%)	0 (0%)	eliminated (POTT handles)
Avg Zone 1 trip (ms)	18	19	+1 ms only

The 2 residual overreach cases: Both occur at $k_{ibr} = 0.5$, $\alpha = 0.95$, $\beta = -90^\circ$ — the extreme corner of the operating space where the IBR shift exceeds the adaptive correction’s update bandwidth (10 ms update vs 8 ms fault transient). These cases are cleared by POTT within 32 ms.

The +1 ms speed penalty: The LM estimator requires 2–3 samples (10–15 ms) to converge to $\hat{k}_{ibr}$. During this estimation period, Zone 1 asserts only after the first adaptive boundary update. This is the source of the 1 ms additional delay.

Coverage Map: TR-35 Figure 2

The full coordination study (TR-35) maps the (k_{ibr}, α) plane into four protection regions:

Region	Condition	Primary function	Time
A	$k_{ibr} < 0.06$	87L (line differential) only	8–20 ms
B	$0.06 \leq k_{ibr}$ and $\alpha \leq 0.80$	Zone 2/POTT + 87L	32 ms
C	$\alpha \leq 0.80$, $k_{ibr} \leq 0.12$	Zone 1 + TDCS	20 ms
D	Remainder covered	Zone 1 + 87L	20 ms

Key result from TR-35: Zero coverage gaps across all 48 sampled (α, k_{ibr}) cells. Every point in the operating space is covered by at least one protection function.

Best Figure to Show at the TSA Meeting

Recommended Figure: TR-35 Figure 2 — Full Protection Coverage Map

What it shows: The (k_{ibr}, α) plane divided into colour-coded regions (A, B, C, D) showing which combination of protection functions covers each operating point. The 87L channel (green band, $k_{ibr} \geq 0.06$) runs in parallel with all distance elements and provides the universal back-stop.

Why it is the best figure:

1. It is the *synthesis* result of Chapters 7–11: no single protection element covers the full plane; the coordinated set does.
2. It shows *zero gaps* — a clear, definitive result.
3. It directly answers “what happens at high k_{ibr} ?” and “what happens for far-end faults?” simultaneously.
4. Examiners can read the regions directly without requiring detailed knowledge of any individual algorithm.

Where to find it: TR-35/2026, page 3, Figure 2. Plot source: SAMBP test suite, coordination matrix module.

Secondary figure: TR-34 Figure 1 (R-X impedance plane with Quad Zone 1/Zone 2 rectangles, Mho circles, and IBR arc fault trajectory). Use this to explain *why* the Mho misses resistive faults and why the quadrilateral is adopted. The IBR fault locus (red dashed line) running along the R -axis outside the Mho circle is immediately intuitive.

Whiteboard supplement: Draw the R - X plane with: (1) a fixed Mho circle, (2) a fixed quadrilateral box, (3) an arrow showing the IBR lagging-injection shift vector, and (4) the *adapted* quadrilateral box after SAMBP correction. This takes 60 seconds and makes the adaptation mechanism visually obvious.

Cross-Thesis Connections

How This Chapter Connects to Other Work

Depends on (upstream):

- **Ch 2 (IBR Characterisation):** The failure modes F4 (overreach), F5 (underreach), F6 (directional loss) that this chapter solves are identified in Ch 2.
- **Ch 6 (Digital Twin / LM Estimator):** The real-time $\hat{k}_{ibr}$ estimate used in Eq. (7) comes from the EKF+LM digital twin. RMSE of $|\alpha - \hat{\alpha}| \leq 0.011$ (Ch 8) bounds the residual correction error.
- **Ch 8 (Cybersecurity):** The GOOSE permissive signal in the POTT scheme is the attack surface addressed in Ch 9 (HMAC-SHA256 authentication, GOOSE replay protection).

Enables (downstream):

- **Ch 7 (WAPC):** The POTT scheme's GOOSE infrastructure is reused by the Wide-Area Protection Coordinator for N-1 contingency coordination.
- **Ch 8 (Cybersecurity):** The GOOSE authentication overhead (0.36 ms) is compatible with the POTT latency budget (≤ 4 ms GOOSE, 8 ms total).
- **TR-35:** This chapter's outputs (Zone 1/2 settings, POTT logic) feed directly into the full coordination study.

SAMBP framework role: This is Contribution C5 (quadrilateral + IBR correction) and C6 (Monte Carlo reliability framework). Both feed into the SAMBP Stage-2 veto gate: if the adaptive reach correction cannot converge within one update cycle, Stage-2 blocks the Zone 1 trip and hands off to POTT (Zone 2). This is the SAMBP *tiered response* — attempt fast Zone 1, escalate to POTT, final backup via 87L.

Key Equations Reference Card

Equations You Must Be Able to Write and Explain

1. Apparent impedance with IBR infeed (slide equation):

$$Z_{app} = \alpha Z_L + Z_F + Z_F k_{ibr} \frac{I_r}{I_A} e^{j\beta}$$

Say: " ΔZ_{ibr} is the IBR-induced shift. When $\beta = -90^\circ$ (lagging Volt-VAr), the shift

has a negative imaginary component, pulling Z_{app} into Zone 1 for faults beyond the 80% reach boundary.”

2. Reactive shift:

$$\Delta X = Z_F k_{\text{ibr}} \frac{I_r}{I_A} \sin \beta$$

Say: “For $\beta = -90^\circ$, $\Delta X < 0$ — the relay sees the fault as closer than it is. This is overreach.”

3. Adaptive reach correction (SAMBP-21):

$$X_{\text{reach,adapt}}(t) = X_{\text{reach,nom}} - k_X \hat{k}_{\text{ibr}}(t) \frac{I_{\text{rated}}}{I_A}$$

Say: “ $k_X \approx 10 \Omega$ is the maximum reactive correction coefficient. We subtract the expected shift before it causes a false trip.”

4. Quadrilateral trip condition:

$$T_{21} = \mathbf{1}[C_X \cap C_R \cap C_{\text{load}}]$$

Say: “Independent X and R reach boundaries, plus a load blinder. The Mho cannot do this.”

5. Resistive reach with IBR infeed amplification:

$$R_{\text{app}} = R_F \left(1 + \frac{1}{k_{\text{ibr}}} \right)$$

Say: “At $k_{\text{ibr}} = 0.1$, a 0.004 pu arc resistance appears as 0.044 pu to the relay. The quadrilateral forward reach is set to cover this.”

6. IBR-secure POTT logic:

$$T_{\text{POTT}} = T_{Z2,A} \cap T_{Z2,B} \cap \mathbf{1}[k_{\text{ibr}} < 0.45]$$

Say: “For $k_{\text{ibr}} > 0.45$, POTT is inhibited and Zone 2 reverts to a 100 ms timer with 110% reach to prevent through-fault false permissives.”

7. POTT end-to-end trip time:

$$t_{\text{total}} = t_{\text{GOOSE}} + t_{\text{CT}} + t_{\text{meas}} \leq 4 + 2 + 2 = 8 \text{ ms}$$

Say: “32 ms on the slide includes a conservative GOOSE latency and relay processing time. The IEC 61850 P1 class guarantees ≤ 4 ms GOOSE transfer.”

Speaker Script (Timed to 3 Minutes)

What to Say at the TSA Meeting

[0:00–0:30] Open with the problem:

“Distance protection is the backbone of transmission line protection worldwide — it measures impedance rather than current, making it inherently selective. But IBR completely changes the measurement. When a Type-4 inverter injects reac-

tive current during fault ride-through, it adds a complex impedance shift ΔZ_{ibr} to whatever the relay measures. For lagging reactive injection — which is exactly what Volt-VAr control does during a nearby voltage dip — this shift pulls the apparent impedance into Zone 1 for faults beyond the 80% reach boundary. The relay declares an instantaneous trip for a fault that is not in its zone. That is overreach.”

[0:30–1:15] Explain the solution:

“SMBP-21 addresses this with two mechanisms. First, the adaptive boundary: instead of fixing Zone 1 at 80% of the line, we update the reactive reach boundary every 10 ms based on the real-time $\hat{k}_{ibr}$ estimate from the LM digital twin. The correction is $X_{adapt} = X_{nom} - k_X \hat{k}_{ibr}$. We pre-compensate for the shift before it causes a false trip. Second, when IBR penetration is high — above 50% — Zone 1 adaptation alone is insufficient, because the estimator convergence time approaches the fault transient duration. In that regime, SMBP-21 immediately asserts a Zone 2 POTT permissive over IEC 61850 GOOSE to the remote end, achieving clearance in under 32 ms — versus the conventional 200–300 ms Zone 2 timer.”

[1:15–2:00] Present the results:

“On 50 simulated faults spanning the full IBR operating envelope, the fixed relay had 17 overreach events and 5 underreach events — 44% of faults were misclassified. SMBP-21 reduces overreach to 2 events and underreach to zero. Correct Zone 1 clearance rises from 74% to 96%. The average trip time increases by only 1 ms, from 18 to 19 ms. Speed and selectivity improve simultaneously.”

[2:00–2:30] Point to TR-35 coverage map:

“The full coordination study in TR-35 maps the (k_{ibr}, α) operating space into four protection regions. The key result is that every point in the space — every combination of IBR penetration and fault location — is covered by at least one protection function. There are zero gaps. Below $k_{ibr} = 0.06$, the 87L line differential is the primary element. Between 0.06 and 0.12, Zone 1 with TDCS. Above 0.12, Zone 2/POTT plus 87L. The coloured map summarises the entire distance protection coordination result in a single figure.”

[2:30–3:00] Close with the contribution:

“Contributions C5 and C6: the IBR-corrected quadrilateral eliminates overreach without sacrificing resistive fault coverage; the Zone 2 POTT with IBR inhibit provides sub-cycle clearance for high-penetration scenarios. Together, these restore distance protection reliability in IBR-dominated networks to the same standard expected in synchronous networks. The Monte Carlo reliability study in C6 quantifies this: detection probability 0.993 versus 0.917 for the classical Mho.”

Anticipated Questions and Answers

Q1: Why not simply lower the Zone 1 reach setting to avoid overreach?

Answer: Reducing the Zone 1 reach from 80% to, say, 65% would push the overreach risk boundary back, but at the cost of leaving 35% of the line unprotected by fast instantaneous tripping. All faults in the “gap” (65–80% of the line) would then require Zone 2 clearance (200–300 ms), which exceeds the generator ride-through capability and the IBR memory polarisation window of 100 ms. The adaptive

approach maintains 80% coverage when k_{ibr} is low and adjusts only when IBR penetration is high enough to cause a real overreach risk. It also inherently handles the underreach case (leading injection) by moving the boundary in the opposite direction.

Q2: The POTT requires communication — what if the GOOSE link fails?

Answer: SAMBP-21 includes a GOOSE watchdog: if no heartbeat is received from the remote end within 2 ms (IEC 61850 P1 timeout), the relay reverts to the local Zone 2 timer (200 ms). This is the same fallback as a conventional distance relay — the POTT provides acceleration, but the relay remains functional without it. The cybersecurity study (Ch 12/TR-52) shows that HMAC-SHA256 authentication adds only 0.36 ms overhead to each GOOSE frame, well within the 4 ms latency budget. GOOSE flooding attacks are mitigated by rate limiting at the switch layer.

Q3: How does the relay know the IBR commanded angle β ?

Answer: The relay does not need to know β explicitly. The LM digital twin estimates $\hat{k}_{ibr}$ from the measured terminal voltages and currents (the EKF+LM algorithm in Ch 8). The adaptive reach correction uses only $\hat{k}_{ibr}$ and I_{rated}/I_A — it does not require β . The correction coefficient k_X is pre-set conservatively at the worst-case angle ($\beta = -90^\circ$): $k_X = |Z_F^{\max}| = 10 \Omega$. If β is less adverse (e.g., $\beta = 0^\circ$), the correction over-compensates slightly, making Zone 1 slightly conservative but not causing underreach (the residual error is $\Delta X_{\text{residual}} \leq 0.055 \Omega$, negligible).

Q4: What about three-phase faults — do they overreach too?

Answer: Yes, but less severely. For a balanced three-phase fault, the IBR injects balanced reactive current, so the apparent impedance shift is purely reactive (ΔX term only, no ΔR component). The overreach condition is identical to the single-phase case: $\Delta X < 0$ for lagging injection. However, three-phase faults are rarer and typically produce larger differential currents (87L trips faster), so the Zone 1 overreach window is shorter. The 50-fault simulation includes three-phase faults; the adaptive correction handles them identically.

Q5: How is Zone 2 POTT different from Distance Protection with pilot wires?

Answer: Classic pilot-wire protection uses dedicated copper wires or a dedicated fibre channel for a differential current comparison signal. POTT uses the existing IEC 61850 substation LAN (GOOSE) — no dedicated channel is needed. GOOSE is a multicast Ethernet frame with sub-4 ms guaranteed latency on a P1-class substation LAN. The POTT logic is entirely implemented in software within the distance relay, making it retrofit-compatible with existing IEC 61850 substations. The key difference from a pilot differential scheme is that POTT is a *permission* signal (“End A has seen a Zone 2 fault”), not a measurement signal — it does not require time-synchronised current samples.

Chapter 5 One-Line Summary for Examiners

“IBR reactive injection distorts apparent impedance by $\Delta Z_{ibr} = Z_F k_{ibr} e^{j\beta}$. SAMBP-21 pre-compensates with a real-time adaptive Zone 1 boundary updated every 10 ms from the LM estimate of $\hat{k}_{ibr}$, and accelerates clearance via POTT when $k_{ibr} > 0.5$. Overreach falls from 34% to 4%; correct Zone 1 trips rise from 74% to 96%; speed penalty is 1 ms. TR-35 shows zero coverage gaps across the full (α, k_{ibr}) operating space.”